Lu Fang

Email: lf2819@tc.columbia.edu | Tel: 646-519-0222

EDUCATION BACKGROUND

Teachers College, Columbia University **09/2024 – 12/2025**
MS in Learning Analytics
Zhejiang University **09/2022 – 06/2024**
MEd in Science and Technology Education **GPA**: 3.96/4.0
Jiangnan University **09/2018 – 06/2022**
BS in Educational Technology **GPA**: 3.7/4.0

PUBLICATION

JOURNAL ARTICLES (UNDER REVIEW)

Zhang, L., **Fang, L.**, Huang, Y., & Shang, J. (2025). Integrating AI tutor into digital game-based learning for Chinese as a foreign language: Impacts on language acquisition, motivation, and learning behaviors. *Manuscript submitted to Education and Information Technologies*.

Zhang, L., Shang, J., **Fang, L.**, & Huang, Y. (2025). An investigation of metacognitive regulation of CFL learners in contextualized digital game-based learning: Analyzing error modification behaviors using lag sequential analysis. *Manuscript submitted to System*.

JOURNAL ARTICLES (ACCEPTED)

(in-press) **Fang, L.**, Tang, G., & Zhang, L. (2025). Helpful or Harmful? A Comparative Study of Perceived and Actual Effectiveness of LLM-Driven Tutors in Game-Based CFL Learning. *Education Sciences*

Zhang, L., **Fang, L.**, & Shang, J. (2024). The application of gamified virtual environments in international Chinese education: Perspectives on discourse cognition and emotional motivation. *E-education Research.*

Fang, L., & Zhai, J. (2023). Promoting the spirit of scientists in science museums: Challenges and opportunities. *Studies on Science Population*.

Qian, Y., & **Fang, L.** (2021). Project-driven instructional design for Python programming: A case study of using the Monte Carlo method to calculate pi. *China Information Technology Education*, 3, 39–40, 92.

CONFERENCE PROCEEDINGS

Feng, M., **Fang, L.**, Yang, G., Wang, S., Wei, Z., Shang, P., & Zhang, L. (2024). How does motivation affect gamified learning engagement in international Chinese education? Empirical research based on lag sequential analysis. In *Proceedings of the 28th Global Chinese Conference on Computers in Education (GCCCE 2024)*. (*Best Chinese Paper Nomination*)

Cheng, Y., Zhang, L., & **Fang, L.** (2023). A study on the design of online virtual contexts for international Chinese language education by integrating large models. In *Proceedings of the Learning Science Research Branch of Chinese Society of Higher Education Academic Conference*.

Fang, L., & Guo, Y. (2023). Research overview on museum mobile learning apps. In *Proceedings of the 18th Annual China Information Technology Education Conference*.

RESEARCH EXPERIENCE

AI-Enhanced Contextualized Game Development for Chinese as foreign language learners **08/2023-present**
Team Leader*, PI: Prof. Junjie Shang* **Learning Science Laboratory** **Peking University**

- Led the development of Ethan's Life in Beijing, a Unity3D-based educational game for Chinese as a Foreign Language (CFL) learners.
- Integrated the iFlytek Spark LLM to create an AI tutor providing real-time feedback.
- Applied K-means clustering and Lag Sequential Analysis to identify and compare learners' metacognitive patterns.
- Collected learner–AI tutor chat logs and analyzed the AI tutor's perceived usefulness using the Technology Acceptance Model (TAM) framework.

User Experience and Usage Characteristics of AITA: An AI Tutor Designed for an Advanced Engineering Programming Course **09/20224-12/2024**
Research Team Member, *PI: Dr. Shu-Yi Hsu* **Columbia University**

- Conducted descriptive analysis of over 30 graduate students' chat histories with AITA, examining usage timing, frequency, conversation topics, and potential underlying factors.
- Conducted semi-structured interviews with over 10 graduate students; analyzed transcripts to identify patterns and characteristics in AITA usage.

Modeling Computational Thinking Skill Through Log Data from Python Learning Platform **09/2021-04/2022**
Independent researcher*, PI: Prof. Yizhou Qian* **Jiangnan University**

- Extracted and aligned high school learners' log data from a Python programming game-based learning platform, mapping coding behaviors to computational thinking (CT) constructs.
- Built a random forest regression model to predict learners' mastery of CT concepts, achieving a 77.69% fit, and conducted feature importance analysis to identify critical coding behaviors influencing learning outcomes.
- Applied *K*-means clustering to categorize students into three distinct CT development profiles, based on patterns in programming practices and behavioral engagement within the platform.

PROFESSIONAL EXPERIENCE

Zhejiang Science and Technology Museum | Intern, Education Department **06/2023–09/2023**

- Designed and facilitated five public science workshops, serving over 200 visitors.
- Produced science-themed animations for the official TikTok channel.

Beijing TAL Co. | Instructional Design Intern **03/2021-09/2021**

- Collaborated on curriculum and materials for children's online programming courses.
- Created educational animations in Scratch for public release.

Wuxi First Middle School | IT Teaching Intern **09/2020–03/2021**

- Designed and taught IT curriculum to two high school classes (~90 students).
- Managed classroom logistics and organized extracurricular STEM activities.

Wuxi Xinqihang Education Technology Co. | Co-founder **12/2018-06/2022**

- Led a team of 20+ educators serving over 1,000 students.
- Designed and implemented 36 Engligh lesson plans for Grade1-3 students.
- Oversaw recruitment, training, and quality control of all instructors.

EXTRACURRICULAR ACTIVITY

Application of C-STEM Courses for Elementary School Students Based on Scratch **05/2020-06/2021**

- Designed scratch-based computational thinking + math courses (~20 courses) and textbooks (~2 textbooks).
- Delivered 30+ volunteer programming sessions to underserved rural students.

AWARD AND HONOR

- National Scholarship for Undergraduate Students (Top 0.1%) | 2019
- 3rd Prize, Jiangsu Teaching Skills Competition for Normal University Students | 2021 (Top 1%)
- National 3rd Prize, 12th User Experience Design Award | 2020 (Top 1%)
- 3rd Prize, Jiangsu Province E-Commerce Innovation Challenge | 2020 (Top 1%)

PROFESSIONAL SKILL

Programming & Analysis: Python, C#, Java, JavaScript, MySQL, SPSS
Languages: Chinese (Native), English (Fluent), Japanese (Intermediate)